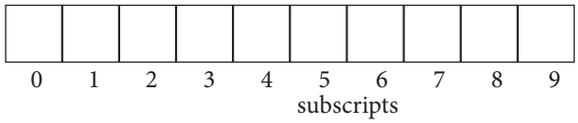
In the above declaration, an array named "num" is declared with 10 elements (memory space to store 10 different values) as integer type.

For the above declaration, the compiler allocates 10 memory locations (boxes) referenced by a common name "num" as given below

int num[10];



Each element (Memory box) has a unique index number starting from 0 which is known as "subscript". The subscript always starts with 0 and it should be an unsigned integer value. Each element of an array is referred by its name with subscript index within the square bracket. For example, num[3] refers to the 4th element in the array.

Some more array declarations with various data types:

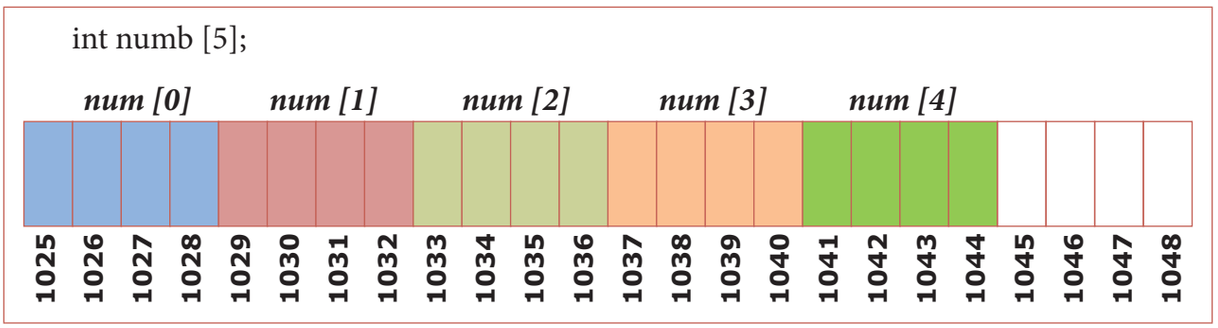
char emp_name[25]; // character array named emp_name with size 25

float salary[20]; // floating-point array named salary with size 20

int a[5], b[10], c[15]; // multiple arrays are declared of type int

Memory representation of an one dimensional array

The amount of storage required to hold an array is directly related with type and size. The following figure shows the memory allocation of an array with five elements.



The above figure clearly shows that, the array num is an integer array with 5 elements. As per the Dev-C++ compiler, 4 bytes are allocated for every int type variable. Here, there are totally 5 elements in the array, where for each element, 4 bytes will be allocated. Totally, 20 bytes will be allocated for this array.

Datatype	Turbo C++	Dev C++
char	1	1
int	2	4
float	4	4
long	4	4
double	8	8
long double	10	10

The memory space allocated for an array can be calculated using the following formula:

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